

Math 526, F06
Quiz 5

Name:
SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices, $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$ are column n -vectors and $\mathbf{A} = \mathbf{B} - \mathbf{u}\mathbf{v}^t$. Then under an appropriate condition $\mathbf{A}$ is invertible and $\mathbf{A}^{-1}$ is given by the **Sherman-Morrison formula**:

$$\mathbf{A}^{-1} = \left[\mathbf{I} + \left(\frac{1}{1 - \mathbf{v}^t(\mathbf{B}^{-1}\mathbf{u})} \right) (\mathbf{B}^{-1}\mathbf{u})\mathbf{v}^t \right] \mathbf{B}^{-1}.$$

Now we are given $\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -2 \\ -1 \\ -3 \\ 4 \end{bmatrix}$.

a) Let $\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}$, find corresponding $\mathbf{u}$ and $\mathbf{v}$ such that $\mathbf{A} = \mathbf{B} - \mathbf{u}\mathbf{v}^t$ and then show that $\mathbf{A}$ is invertible.

Solution:

$$\mathbf{A} = \mathbf{B} - \begin{bmatrix} 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \mathbf{B} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot [0 \ 0 \ -1 \ -1]$$

that means

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ -1 \end{bmatrix}.$$

Then we solve

$$\mathbf{B}\mathbf{z} = \mathbf{u}$$

and find

$$\mathbf{z} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 2 \end{bmatrix}$$

So,

$$\gamma = 1 - \mathbf{v}^t\mathbf{z} = 1 - (-1) = 2 \neq 0$$

that means $\mathbf{A}$ is nonsingular.

b) Use the information from (a) to solve the system $\mathbf{Ax} = \mathbf{b}$ by the algorithm based on the Sherman-Morrison formula.

Solution: We solve

$$\mathbf{By} = \mathbf{b}$$

and find

$$\mathbf{y} = \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

According to the Sherman-Morrison formula, we have

$$\begin{aligned} \mathbf{x} &= \mathbf{A}^{-1}\mathbf{b} \\ &= \mathbf{y} + \frac{\mathbf{v}^t\mathbf{y}}{\gamma}\mathbf{z} \\ &= \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \end{bmatrix} + \left(\frac{0}{2}\right) \cdot \begin{bmatrix} 1 \\ -1 \\ -1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$